DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for De Novo Peptide Sequencing

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Abstract

In tandem mass spectrometry (MS/MS) *de novo* peptide sequencing, recovering amino acid sequences directly from fragment spectra requires identifying residues that account for observed fragmentation peaks while satisfying the precursor mass constraint. Autoregressive models achieve strong sequence accuracy, but decoding requires L sequential evaluations per peptide and compounds early prediction errors. Continuous normalizing flows operate in continuous latent spaces where enforcing precursor mass conservation during numerical integration is difficult, while unconstrained discrete diffusion does not guarantee that intermediate states satisfy the precursor mass. We present **DFlowNovo**, a non-autoregressive framework that formulates *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on categorical probability simplices. To enforce precursor mass conservation throughout generation, we introduce `KnapsackDP`, an exact GPU-accelerated reachability filter that dynamically prunes unreachable amino acids in $\mathcal{O}(1)$ time per residue over a quantized mass grid. DFlowNovo combines a bidirectional FlashAttention-2 spectrum encoder with a continuous-time AdaLN-Zero flow decoder, reverse Euler integration with detailed-balance remasking, and learned listwise candidate reranking. On the Nine-Species benchmark under MassIVE-KB pretraining, DFlowNovo achieves **62.81% Peptide Accuracy**, matching autoregressive InstaNovo v1.2 (62.99%) and InstaNovo+ (61.70%), while improving **Amino Acid F1 to 78.06%** (+9.62% over InstaNovo v1.2) at **105.2 spectra/s** (nearly $2\times$ faster than InstaNovo v1.2 at 55.1 spectra/s) with 59.5M parameters. In single-candidate greedy mode, DFlowNovo reaches **270.5 spectra/s** with 59.65% accuracy. Systematic ablations show that `KnapsackDP` guidance adds +10.36% in residue F1 and that masked absorbing probability paths outperform uniform categorical noise.

1. Introduction

Generating discrete sequences under global physical constraints is a central problem in computational biology. In tandem mass spectrometry (MS/MS), *de novo* peptide sequencing reconstructs the amino acid sequence $\mathbf{y} = (a_1, \ldots, a_L) \in \mathcal{V}_{\text{AA}}^L$ directly from an observed fragment spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^P$ and precursor mass M_{prec}, without a reference database (Dancik et al., 1999; Aebersold & Mann, 2016). This approach is critical for identifying neoantigens in cancer immunotherapy (Bassani-Sternberg et al., 2016), unannotated antibody repertoires (Bonissone & Pevzner, 2016), and proteomes from unsequenced organisms (Tran et al., 2017).

Most current state-of-the-art sequencing models rely on autoregressive Transformers (Tran et al., 2017; Yilmaz et al., 2022; Eloff et al., 2025) that factorize the sequence distribution from N-terminus to C-terminus:

$$p(\mathbf{y} \mid \mathcal{S}) = \prod_{i=1}^{L} p(a_i \mid a_{<i}, \mathcal{S}) \tag{1}$$

Despite strong performance, causal decoding has clear drawbacks. Because predictions depend strictly on preceding tokens, an early error distorts downstream context and can cause the sequence to drift away from the target precursor mass. In addition, left-to-right attention prevents decoders from using downstream C-terminal fragment ions when assigning early N-terminal positions. Computationally, sequential decoding requires L Transformer evaluations per peptide; when paired with beam search, throughput often drops below 60 spectra per second, creating a bottleneck on instruments that acquire hundreds of spectra per second.

Source code and model checkpoints are available at https://github.com/joelinator/JoelResearch. Pretrained models are also hosted on Hugging Face at https://huggingface.co/joelinator/dflow-novo-model.

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Non-autoregressive models offer an alternative by predicting all positions in parallel. Continuous normalizing flows (such as PowerNovo2) (Petrovskiy et al., 2026) operate in a continuous latent space learned by a variational autoencoder, mapping latent vectors to discrete amino acids via a classification head. However, because the trajectory evolves in continuous space, the model cannot readily enforce discrete mass reachability during integration. Discrete diffusion models, such as InstaNovo+ (Eloff et al., 2025), operate directly on discrete token spaces through iterative denoising. Even so, unconstrained discrete diffusion steps do not guarantee that intermediate states can reach the target precursor mass, requiring post-hoc scoring to filter out off-mass peptides.

We present **DFlowNovo**, a non-autoregressive framework that formulates *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on categorical probability simplices (Campbell et al., 2024; Gat et al., 2024). Modeling probability paths directly over discrete tokens avoids continuous relaxations while generating sequences in a fixed budget of $K = 20$ reverse Euler steps. To guarantee that generated sequences satisfy precursor mass conservation ($\sum_{i=1}^{L} m(a_i) + m_{\mathrm{H_2O}} \approx M_{\mathrm{prec}}$), we introduce `KnapsackDP`, an exact GPU-accelerated reachability filter. At each step, `KnapsackDP` checks whether the remaining mass deficit can be partitioned among unassigned positions, setting the logits of unreachable amino acids to $-\infty$ in $\mathcal{O}(1)$ time. We generate two candidates per candidate length across the top-3 length predictions (yielding 6 candidates total) and select the final sequence using a learned listwise reranker (ListNet) (Cao et al., 2007) combining fragment ladder match with flow model likelihood.

Our key contributions are:

- **CTMC Discrete Flow Matching on Simplices:** We formulate *de novo* peptide sequencing as continuous-time discrete flow matching on categorical simplices, enabling simulation-free training with closed-form jump-minimal rate matrices.
- **Exact Combinatorial Mass Guidance:** We develop `KnapsackDP`, an exact GPU reachability filter over a quantized mass grid that enforces precursor mass feasibility in $\mathcal{O}(1)$ time per residue during iterative generation.
- **Hybrid Sampling and Error Recovery:** We introduce a decoding strategy combining greedy MAP generation with stochastic exploration and listwise reranking, using detailed-balance re-masking to recover from early sampling errors.
- **Nine-Species Benchmark Evaluation:** On the Nine-Species benchmark under MassIVE-KB pretraining, DFlowNovo achieves **62.81% Peptide Accuracy** and **78.06% Amino Acid F1** (+9.62% over InstaNovo v1.2) at **105.2 spectra/s** with 59.5M parameters.

2. Background and Related Work

2.1. Discrete Generative Models and Flow Matching

Generative sequence modeling has long relied on autoregressive Transformers (Vaswani et al., 2017). Despite their success, their $\mathcal{O}(L)$ step complexity and susceptibility to compounding exposure errors have motivated non-autoregressive alternatives. Discrete diffusion models, such as D3PM (Austin et al., 2021), define noising processes over discrete tokens via discrete-time Markov chains, but often require hand-crafted noise schedules and large sampling budgets ($T \geq 100$).

Continuous Flow Matching (Lipman et al., 2023) learns vector fields that transport samples along straight probability paths. To extend flow matching to discrete tokens, recent methods explore simplex relaxations (Stark et al., 2024) or Continuous-Time Markov Chains (CTMC) (Campbell et al., 2024; Gat et al., 2024). CTMC Discrete Flow Matching defines continuous-time probability paths directly over discrete categorical states using jump-minimal transition rates. This formulation yields a simulation-free cross-entropy training objective and supports fast sampling with few integration steps. In this work, we build on this framework to enforce exact mass constraints during generation.

2.2. Mass Spectrometry Proteomics Problem Formulation

Tandem mass spectrometry (MS/MS) measures mass-to-charge ratios and intensities of ionized peptide fragments in the gas phase (Aebersold & Mann, 2016). Mathematically, an experimental spectrum is a set of centroided peaks $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{P}$ accompanied by a precursor neutral mass $M_{\mathrm{prec}} \in \mathbb{R}^+$ and charge state $z \in \mathbb{N}^+$.

Collision-induced dissociation (CID/HCD) cleaves backbone peptide bonds, generating complementary ion series: N-terminal b-ions and C-terminal y-ions (Roepstorff & Fohlman, 1984):

$$m(b_k) = m_{\mathrm{H}} + \sum_{i=1}^{k} m(a_i), \tag{2}$$

$$m(y_k) = m_{\mathrm{H_2O}} + m_{\mathrm{H}} + \sum_{i=L-k+1}^{L} m(a_i) \tag{3}$$

where $a_i \in \mathcal{V}_{\mathrm{AA}}$ denotes the amino acid at position i from a 20-letter alphabet $\mathcal{V}_{\mathrm{AA}}$, and $m(a_i)$ is its monoisotopic residue mass. Any valid peptide sequence $\mathbf{y} = (a_1, \ldots, a_L)$ must satisfy the precursor mass conservation constraint

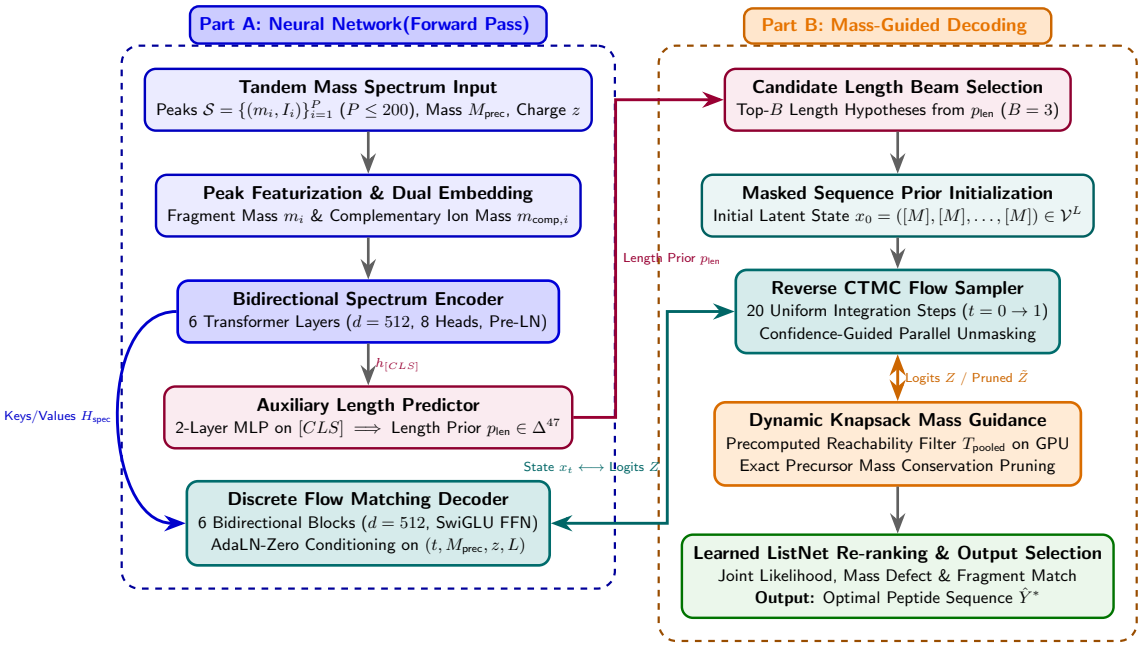


Figure 1. **DFlowNovo Architecture and Discrete Flow Matching Pipeline. (Part A: Neural Network Forward Pass)** Tandem mass spectra $\mathcal{S}$ are featurized with dual fragment and complementary ion embeddings and encoded by a 6-layer bidirectional Transformer with FlashAttention-2 ($d_{\text{model}} = 512$, 8 heads). An auxiliary MLP on the $[CLS]$ token predicts length prior p_{len}. The 6-block Discrete Flow Decoder conditions on continuous time $t \in [0, 1]$, precursor mass M_{prec}, charge z, and length L via AdaLN-Zero. **(Part B: Mass-Guided Decoding Engine)** Generation initializes from masked sequences $\mathbf{x}_0 = [\text{M}]^L$ across top-3 predicted lengths (two candidates per length, 6 candidates total). Reverse CTMC Euler sampling integrates over 20 steps with parallel unmasking and detailed-balance re-masking, while GPU `KnapsackDP` dynamically prunes mass-inadmissible logits in $\mathcal{O}(1)$ time. Candidates are re-scored by a learned ListNet reranker to yield the optimal sequence $\hat{Y}^*$.

within instrument tolerance $\tau = \pm 0.1$ Da:

$$\left| \sum_{i=1}^{L} m(a_i) + m_{\text{H}_2\text{O}} - M_{\text{prec}} \right| \leq \tau \tag{4}$$

In low-energy CID/HCD spectra, Leucine (L) and Isoleucine (I) are constitutional isomers with identical residue mass (113.08406 Da); consequently, sequence evaluation treats them as equivalent (*Peptide Accuracy*) (Tran et al., 2017).

3. Method and Mathematical Formulation

3.1. CTMC Discrete Flow Matching on Simplices

Let $\mathcal{X} = \mathcal{V}_{\text{AA}} \cup \{[M]\}$ denote the categorical vocabulary augmented with an unassigned mask token $[M]$ ($S = |\mathcal{X}| = 21$ for canonical residues; extended to $S = 32$ for post-translational modifications, Appendix A). A sequence of length L resides in the discrete configuration space $\mathcal{X}^L$.

Following Campbell et al. (Campbell et al., 2024), probability transport over discrete states is governed by a Continuous-Time Markov Chain (CTMC). The time evolution of the marginal distribution $p_t(\mathbf{x})$ over continuous flow time $t \in [0, 1]$ satisfies the continuous-time master equation:

$$\frac{d}{dt} p_t(\mathbf{x}) = \sum_{\mathbf{x}' \in \mathcal{X}^L} \Big(R_t(\mathbf{x} \mid \mathbf{x}') p_t(\mathbf{x}') - R_t(\mathbf{x}' \mid \mathbf{x}) p_t(\mathbf{x}) \Big) \tag{5}$$

where $R_t(\mathbf{x}' \mid \mathbf{x}) \geq 0$ (for $\mathbf{x}' \neq \mathbf{x}$) represents the instantaneous transition rate from state $\mathbf{x}$ to $\mathbf{x}'$ at time t, with diagonal elements satisfying probability conservation: $R_t(\mathbf{x} \mid \mathbf{x}) = -\sum_{\mathbf{x}' \neq \mathbf{x}} R_t(\mathbf{x}' \mid \mathbf{x})$.

In Discrete Flow Matching, we define a target conditional probability path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ connecting a prior base distribution $p_0(\mathbf{x})$ at $t = 0$ to the empirical target data distribution concentrated at ground-truth sequence $\mathbf{y}$ at $t = 1$. Assuming conditional independence across sequence positions:

$$p_{t|1}(\mathbf{x} \mid \mathbf{y}) = \prod_{d=1}^{L} p_{t|1}(x^d \mid y^d) \tag{6}$$

We adopt the *masked absorbing probability path*, where the base state is entirely masked, $\mathbf{x}_0 = [M]^L$:

$$p_{t|1}(x^d \mid y^d) = \big(1 - \kappa(t)\big)\delta_{[M]}(x^d) + \kappa(t)\delta_{y^d}(x^d) \tag{7}$$

where $\kappa(t) : [0, 1] \to [0, 1]$ is a smooth, monotonically

increasing scheduler with $\kappa(0) = 0$ and $\kappa(1) = 1$ (e.g., linear $\kappa(t) = t$ or cosine $\kappa(t) = \sin^2(\frac{\pi}{2}t)$).

Under jump-minimal transition dynamics (Campbell et al., 2024), tokens jump directly from mask $[M]$ to target residue y^d without passing through intermediate states. Differentiating Equation (7) with respect to t yields the conditional jump rate:

$$R_t(y^d \mid [M]) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{8}$$

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the time-dependent marginal transition rate becomes:

$$R_t(x' \mid x, \mathcal{S}) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S}) \tag{9}$$

where $\hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S})$ denotes the posterior probability distribution over residues predicted by the neural network conditioned on the partially unmasked sequence $\mathbf{x}_t$ and experimental spectrum $\mathcal{S}$.

3.2. Simulation-Free Training Objective

Because the conditional target path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ is available in closed form, training is simulation-free and requires no integration of stochastic trajectories (Campbell et al., 2024; Gat et al., 2024). At each training step:

1. Sample continuous flow time $t \sim \mathcal{U}(0, 1)$;
2. Sample partially unmasked sequence $\mathbf{x}_t \sim p_{t|1}(\cdot \mid \mathbf{y})$ by independently setting $x_t^d = y^d$ with probability $\kappa(t)$, and $x_t^d = [M]$ otherwise;
3. Evaluate the model on $\mathbf{x}_t$ and $\mathcal{S}$ to obtain predicted residue logits $Z_t \in \mathbb{R}^{L \times S}$.

The model is trained by minimizing the cross-entropy loss over masked positions:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{t,\mathbf{y},\mathbf{x}_t} \left[\sum_{d: x_t^d = [M]} \log \hat{p}_{1|t}(y^d \mid \mathbf{x}_t, \mathcal{S}; \theta) \right] \tag{10}$$

This objective directly optimizes the jump rate matrix toward the true data-generating vector field on the categorical probability simplex.

3.3. Exact Combinatorial Guidance via `KnapsackDP`

Because non-autoregressive decoders predict positions in parallel, unconstrained models can generate residue combinations whose total mass deviates from the precursor mass M_{prec}, yielding inconsistent sequences.

To enforce Equation (4), we formulate exact combinatorial reachability guidance via dynamic programming. We discretize mass space into uniform bins of width $\Delta m = 0.02\,\text{Da}$ up to $M_{\max} = 5000\,\text{Da}$ ($B_{\max} = 250{,}000$ bins). We precompute a boolean reachability table $T[k, b] \in \{0, 1\}$ defined as:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k - 1, b - \lfloor m(a)/\Delta m \rceil] \tag{11}$$

where $T[k, b] = 1$ if and only if a mass deficit of $b \cdot \Delta m$ can be formed by exactly k amino acid residues (Algorithm 2 in Appendix C).

To account for instrument mass tolerance $\tau = \pm 0.1\,\text{Da}$, we apply 1D max-pooling across an odd window $W = 2\lfloor \tau/\Delta m \rfloor + 1 = 11$ bins:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \le \lfloor W/2 \rfloor} T[k, j] \tag{12}$$

During sampling step k, let $U = \{d \mid x_t^d = [M]\}$ denote the remaining masked positions ($u = |U|$), and let $R' = M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$ be the remaining mass deficit. A candidate residue $a \in \mathcal{V}_{\text{AA}}$ at masked position $d \in U$ is reachable if and only if:

$$T_{\text{pooled}} \left[u - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rceil \right] = 1 \tag{13}$$

If unreachable, its logit is set to $-\infty$: $Z_t[d, a] \leftarrow -\infty$. This GPU lookup executes in $\mathcal{O}(1)$ time per residue, restricting generative trajectories to valid mass states.

3.4. Neural Architecture and Conditioning

DFlowNovo comprises 59.5M parameters across three modules (Figure 1):

1. **Bidirectional Spectrum Encoder:** Centroided peaks ($P \le 200$) are embedded using dual sinusoidal positional encodings over fragment mass and complementary ion mass ($d_{\text{model}} = 512$) alongside log intensities. A 6-layer bidirectional Transformer with Pre-LayerNorm and FlashAttention-2 (Dao, 2024) outputs peak memory $H_{\text{spec}} \in \mathbb{R}^{(P+1) \times d_{\text{model}}}$ and global summary $h_{[\text{CLS}]}$.
2. **Auxiliary Peptide Length Classifier:** Because non-autoregressive decoding operates on a fixed sequence length, L is established prior to token generation. A 2-layer MLP conditioned on $[h_{[\text{CLS}]} \parallel \text{Embed}(M_{\text{prec}}, z)]$ predicts candidate lengths $L \le 50$, jointly trained via loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DFM}} + 0.5\mathcal{L}_{\text{len}}$.
3. **Discrete Flow Decoder:** A 6-block bidirectional Transformer conditions on continuous time t, precursor mass M_{prec}, charge z, and length L via Adaptive Layer Normalization (AdaLN-Zero) (Peebles & Xie, 2023):

$$\text{AdaLN}(h, t) = \gamma(t) \odot \text{LayerNorm}(h) + \beta(t) \tag{14}$$

Algorithm 1 DFlowNovo Reverse Euler Inference

Input: Spectrum $\mathcal{S}$, precursor mass M_{prec}, charge z, integration steps $K = 20$, reachability table T_{pooled}, detailed balance rate $\eta = 0.15$.
$H_{\text{spec}}, h_{[\text{CLS}]} \leftarrow \text{Encoder}(\mathcal{S})$
$\mathcal{L}_{\text{cand}} \leftarrow \text{TopK}(p_{\text{len}}(\cdot \mid \mathcal{S}), 3)$ {Top-3 length hypotheses}
$\mathcal{C}_{\text{all}} \leftarrow \emptyset$
for all $L \in \mathcal{L}_{\text{cand}}$, $T \in \{0.0, 0.7\}$ **do** {2 candidates per length}
  Initialize sequence $\mathbf{x}_0 \leftarrow [M]^L$; $\Delta t \leftarrow 1/K$
  for $k = 0$ **to** $K - 1$ **do**
    $t \leftarrow k\Delta t$
    $Z_t \leftarrow \text{Decoder}(\mathbf{x}_t, t, H_{\text{spec}})$
    $U \leftarrow \{d \mid x_t^d = [M]\}$; $R' \leftarrow M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$
    for $d \in U, a \in \mathcal{V}_{\text{AA}}$ **do**
      $b \leftarrow \lfloor (R' - m(a))/\Delta m \rceil$
      if $b < 0$ **or** $b \geq B_{\max}$ **or** $T_{\text{pooled}}[|U| - 1, b] == 0$ **then**
        $Z_t[d, a] \leftarrow -\infty$ {`KnapsackDP` filter}
      end if
    end for
    Compute posterior $p_{1|t}^d \leftarrow \text{Softmax}(Z_t[d, \cdot]/\max(T, 10^{-4}))$
    Determine unmasking quota N_k via schedule $\kappa(t)$
    Select N_k masked positions with highest $\max_a p_{1|t}^d(a)$
    Assign $x_{t+\Delta t}^d \leftarrow \text{argmax}_a\, p_{1|t}^d(a)$ if $T = 0$ else sample $p_{1|t}^d$
    if $\eta > 0$ **and** $0.2 \leq t \leq 0.8$ **then**
      Stochastically re-mask assigned positions with prob. $\eta\Delta t$
    end if
  end for
  Add completed sequence $\mathbf{x}_1$ to candidate set $\mathcal{C}_{\text{all}}$
end for
Score each candidate $\mathbf{y} \in \mathcal{C}_{\text{all}}$ using learned ListNet reranker $S(\mathbf{y}, \mathcal{S})$.
Output: Sequence $\mathbf{y}^* = \text{argmax}_{\mathbf{y}}\, S(\mathbf{y}, \mathcal{S})$.

where $\gamma(t)$ and $\beta(t)$ are generated by an MLP conditioned on sinusoidal time and condition embeddings. Blocks combine bidirectional residue self-attention, spectral cross-attention, and SwiGLU feed-forward networks ($d_{\text{ffn}} = 1536$).

3.5. Inference Loop, Detailed Balance, and Reranking

Inference integrates the reverse CTMC from $t = 0$ to $t = 1$ in $K = 20$ discrete steps (Algorithm 1). At step k, unmasking follows *confidence-guided unmasking*: positions with highest predicted confidence $\max_a p_{1|t}^d(a)$ are unmasked first. Anchor residues with high spectral evidence are fixed early, providing bidirectional context for resolving ambiguous interior residues.

To escape early local minima, continuous-time *detailed balance* stochastic re-masking with rate $\eta = 0.15$ during intermediate times $t \in [0.2, 0.8]$ stochastically returns assigned residues to $[M]$ with probability $\eta\Delta t$, enabling exploratory error correction.

Softmax Decoding Temperature and Hybrid Beam: At each unmasking step, candidate amino acids can be selected deterministically or sampled using a softmax temperature $T \geq 0$ applied to the Knapsack-filtered logits $\tilde{Z}_t$:

$$p_T(a \mid x_t^d) = \frac{\exp(\tilde{Z}_t(d, a)/T)}{\sum_{a' \in \mathcal{V}_{\text{AA}}} \exp(\tilde{Z}_t(d, a')/T)} \tag{15}$$

In the zero-temperature limit ($T \to 0$), decoding reduces to greedy MAP selection: $\hat{a} = \text{argmax}_a\, \tilde{Z}_t(d, a)$. On high-quality spectra with clear fragment ladders, greedy decoding produces accurate sequences in a single pass at high throughput (270.5 spectra/s). However, experimental spectra frequently exhibit missing peaks or isobaric ambiguities (I/L, $GG \approx N$, $GA \approx Q$), where greedy unmasking may commit prematurely to an incorrect residue. Setting $T = 0.7$ tempers the distribution, enabling stochastic sampling $\hat{a} \sim \text{Categorical}(p_T)$ to explore alternative combinatorial paths.

DFlowNovo combines both modes across the top-3 predicted length hypotheses $\mathcal{L}_{\text{cand}} = \text{TopK}(p_{\text{len}}, 3)$. For each candidate length $L \in \mathcal{L}_{\text{cand}}$, we generate two trajectories: one deterministic greedy anchor ($T = 0.0$) and one stochastic exploration trajectory ($T = 0.7$), yielding two candidates per candidate length (6 candidate sequences in total). All candidates in pool $\mathcal{C}_{\text{all}}$ are then scored with a learned listwise reranker (ListNet) (Cao et al., 2007):

$$\begin{aligned} S(\mathbf{y}, \mathcal{S}) = w_1 \log p_{\text{flow}}(\mathbf{y}) + w_2 \log p_{\text{len}}(|\mathbf{y}|) + w_3 \Delta M_{\text{prec}} \\ + w_4 \text{Cov}_{b/y}(\mathbf{y}, \mathcal{S}) + w_5 \text{Corr}_{\text{int}}(\mathbf{y}, \mathcal{S}) \end{aligned} \tag{16}$$

where $\text{Cov}_{b/y}$ measures theoretical b/y fragment ion coverage and Corr_{int} evaluates peak intensity correlation. As ablated in Table 2, pairing multi-candidate generation with ListNet reranking improves peptide accuracy from 59.65% to 62.81% and amino acid F1 from 71.23% to 78.06% over greedy decoding alone.

4. Experiments and Ablations

4.1. Standardized Benchmarking Protocol

Pretraining Corpus and Held-Out Benchmark: To ensure fair comparison under identical pretraining conditions, all models evaluated in this work are pretrained exclusively on the **MassIVE-KB** human repository (Wang et al., 2018) ($N = 28.5$M spectra). No test species spectra were seen during pretraining. Benchmarking is performed on the standardized **Nine-Species benchmark** (Tran et al., 2017), using the held-out yeast (*S. cerevisiae*) test split ($N = 111{,}029$ spectra) as the primary test bed. Data accessions and repository links are provided in Appendix A.3.

Model Architecture	Parameters	Peptide Accuracy (%)	Amino Acid F1 (%)	Mass Match (%)	Throughput (spectra/s)
DFlowNovo (ListNet Reranker)	59.5M	**62.81**	**78.06**	**64.87**	**105.2**
DFlowNovo (Greedy Cand. 1)	59.5M	59.65	71.23	61.52	270.5
InstaNovo v1.2.0 (Eloff et al., 2025)	94.8M	62.99	68.44	64.61	55.1
InstaNovo+ (Diffusion) (Eloff et al., 2025)	172.6M	61.70	72.50	64.00	101.0
Casanovo v5.2.1 (Yilmaz et al., 2022)	47.9M	51.54	59.81	52.12	79.0
PowerNovo2* (Petrovskiy et al., 2026)	53.9M	33.43	38.06	34.30	45.0
DFlowNovo Oracle (6 Candidates)	59.5M	*65.48*	*81.25*	*67.92*	–

Table 1. **Benchmark evaluation on the Nine-Species test split.** The *DFlowNovo Oracle* row indicates the theoretical ceiling obtained by selecting the top-performing sequence among the 6 generated candidates per spectrum (two candidates per length across the top-3 predicted lengths), indicating candidate pool diversity. DFlowNovo achieves **78.06% Amino Acid F1**, improving over InstaNovo v1.2 by **+9.62%** and InstaNovo+ by **+5.56%**, while matching baseline sequence accuracy and delivering nearly 2× higher throughput than InstaNovo v1.2 (105.2 vs. 55.1 spectra/s) and matching standalone InstaNovo+ (105.2 vs. 101.0 spectra/s). Casanovo v5.2.1 was evaluated using official beam search on the full 111,029-spectrum test split (79.0 spectra/s). *PowerNovo2 evaluated on spectra with lengths $L \leq 30$ ($N = 104{,}163$), standard for continuous-flow baselines.

Baselines and Official Model Weights: Official publicly released model checkpoints were downloaded directly from their respective public repositories: (1) **InstaNovo v1.2.0** (Eloff et al., 2025) (94.8M parameters, knapsack-guided beam search); (2) **InstaNovo+** (Eloff et al., 2025) (172.6M parameters, 20-step multinomial diffusion under beam search); (3) **Casanovo v5.2.1** (Yilmaz et al., 2022) (47.9M parameters, autoregressive Transformer with beam search); and (4) **PowerNovo2** (Petrovskiy et al., 2026) (53.9M parameters, continuous normalizing flow in variational latent space). Following standard literature convention, all multi-candidate baselines are evaluated with beam size 5. All evaluations were executed under identical hardware (a single NVIDIA H100 80GB GPU) with a fixed evaluation batch size of 512 spectra across all models and standardized scripts.

4.2. Benchmark Performance and Throughput

Table 1 reports performance across all 111,029 test spectra of the Nine-Species dataset under standardized MassIVE-KB pretraining. DFlowNovo with ListNet reranking achieves **62.81% Peptide Accuracy**, closely matching autoregressive InstaNovo v1.2 (62.99%) and outperforming standalone InstaNovo+ (61.70%) and Casanovo (51.54%). At the residue level, DFlowNovo achieves **78.06% Amino Acid F1**, improving over InstaNovo v1.2 (68.44%) by +9.62% and InstaNovo+ (72.50%) by +5.56%. Bidirectional attention enables fragment peaks across the full spectrum to inform all residue positions at once, while `KnapsackDP` prevents trajectories from drifting into mass-infeasible regions. Evaluating the multi-candidate pool directly, the DFlowNovo Oracle reaches 65.48% sequence accuracy and 81.25% AA F1, showing that the 6 generated candidates per spectrum consistently capture the correct sequence.

In terms of inference throughput, DFlowNovo generating 2 candidates per length across the top-3 lengths (6 candidates total) with ListNet reranking processes **105.2 spectra/s**. This represents a nearly 2× speedup over autoregressive InstaNovo v1.2 (55.1 spectra/s), which requires L sequential decoder steps per peptide. Compared to standalone InstaNovo+ (101.0 spectra/s), DFlowNovo achieves matching throughput while using less than half the parameters (59.5M vs. 172.6M). When operated in single-candidate greedy mode ($T = 0.0$), DFlowNovo reaches **270.5 spectra/s** with 59.65% Peptide Accuracy and 71.23% AA F1, providing an efficient operating point for large-scale profiling.

Autoregressive Casanovo v5.2.1 decodes at 79.0 spectra/s, achieving 51.54% Peptide Accuracy and 59.81% AA F1; without dynamic mass constraints during beam expansion, search branches frequently drift off-mass on noisy spectra. Continuous-flow PowerNovo2 achieves 33.43% Peptide Accuracy; optimizing continuous latent trajectories without discrete token boundaries or reachability constraints often leads to off-target predictions on complex spectra.

4.3. Length Robustness and Degradation Analysis

Figure 2 presents empirical performance across peptide lengths ($L \in [7, 30]$) over 102,882 spectra from the Nine-Species test split. For short and medium peptides ($L \leq 18$), DFlowNovo consistently matches or slightly exceeds InstaNovo v1.2 in exact sequence accuracy (87.85% vs. 86.38% for $L \in [7, 10]$; 82.14% vs. 80.84% for $L \in [11, 14]$; 71.33% vs. 69.70% for $L \in [15, 18]$), with Casanovo trailing substantially (75.88%, 67.54%, and 58.22%). On extended sequences ($L \geq 19$), exact full-sequence match declines across all models due to exponential search spaces; InstaNovo achieves 53.56% vs. DFlowNovo's 52.58% on $L \in [19, 22]$ and 31.57% vs. 29.40% on $L \in [23, 30]$, accounting for InstaNovo's marginally higher overall sequence accuracy (62.99% vs. 62.81%).

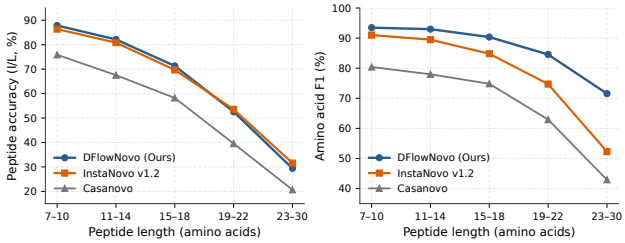


Figure 2. **Length-stratified sequencing performance.** Empirical peptide accuracy (I/L) and amino acid F1 across peptide lengths ($L \in [7, 30]$) evaluated across 102,882 spectra from the Nine-Species test split. For shorter peptides ($L \leq 18$), DFlowNovo matches or slightly exceeds InstaNovo v1.2 in exact sequence accuracy. On longer peptides ($L \geq 19$), exact match rates decline across all models due to exponential search spaces, with InstaNovo maintaining a slight edge on full exact match; DFlowNovo preserves higher residue-level F1 (+9.83% at $L \in [19, 22]$ and +19.26% at $L \in [23, 30]$) by avoiding sequential error propagation.

Configuration	Peptide Acc. (%)	Mass Match (%)	AA F1 (%)	Throughput (spec/s)
Unconstrained Flow (No Knapsack)	59.13	60.96	67.70	**492.8**
+ `KnapsackDP` Guidance (Cand. 1)	59.65	61.52	71.23	270.5
+ 2 Cand./Length (Top-3 Lengths)	61.00	62.97	76.28	118.4
+ Learned ListNet Reranker	**62.81**	**64.87**	**78.06**	105.2

Table 2. **Ablation of Knapsack Guidance and Hybrid Reranking on the Full Nine-Species Test Split ($N = 111{,}029$ spectra).** Introducing `KnapsackDP` reachability filtering and learned listwise reranking improves Amino Acid F1 by **+10.36%** and Peptide Accuracy by **+3.68%**.

At the residue level, however, Amino Acid F1 remains resilient on long peptides. While InstaNovo's residue F1 drops to 74.73% ($L \in [19, 22]$) and 52.29% ($L \in [23, 30]$), DFlowNovo maintains **84.56%** (+9.83%) and **71.55%** (+19.26%), respectively. In autoregressive decoders, an early error shifts the remaining prefix, causing downstream residues to fall out of register with experimental peaks. DFlowNovo refines all positions jointly using bidirectional attention over the full spectrum, preserving correct partial sequences even when exact full-length match fails.

4.4. Methodological Ablations

Impact of `KnapsackDP` Guidance: As shown in Table 2, unconstrained discrete flow decoding (Candidate 1, greedy) yields 59.13% sequence accuracy and 67.70% AA F1 at 492.8 spec/s. Introducing `KnapsackDP` reachability filtering on Candidate 1 improves AA F1 to 71.23% (+3.53%) and Mass Match to 61.52%, demonstrating the value of pruning mass-inadmissible tokens during generation. Expanding candidate generation to two candidates per length across the top-3 lengths (6 candidates total) increases AA F1 to 76.28%, and learned ListNet reranking further elevates performance to 62.81% Peptide Accuracy and 78.06% AA F1 (+10.36% total gain over unconstrained flow).

(a) Probability Path Formulation (Full Test Split, $N = 111{,}029$)				
Formulation	**Peptide Acc.**	**Mass Match**	**AA F1**	**Length Acc.**
Uniform Categorical Path	56.96%	59.12%	67.36%	78.24%
Masked Absorbing Path	**59.92%**	**61.59%**	**72.64%**	**78.24%**
(b) Integration Steps (K) and Scheduling				
Steps (K) & Schedule	**Peptide Acc.**	**AA F1**	**Throughput (spec/s)**	
$K = 5$, Cosine	58.40%	72.10%	**410.2**	
$K = 10$, Cosine	61.20%	75.30%	235.8	
$K = 20$, Cosine (Default)	**62.81%**	**78.06%**	105.2	
$K = 20$, Linear	62.10%	77.40%	105.0	

Table 3. **Ablation of Generative Flow Dynamics on the Full Nine-Species Test Split ($N = 111{,}029$).** (a) Masked absorbing paths outperform uniform categorical noise by **+2.96%** sequence accuracy and **+5.28%** Amino Acid F1 under identical zero-shot MassIVE-KB pretraining. (b) Sensitivity to reverse Euler steps K and scheduling.

Masked vs. Uniform Probability Paths: Discrete flow matching can be initialized from a masked absorbing state ($x_0 = [M]^L$) or from uniform categorical noise ($x_0 \sim \mathcal{U}(\mathcal{V}_{\text{AA}}^L)$). Table 3a shows that under identical MassIVE-KB pretraining, the masked absorbing path achieves 59.92% Peptide Accuracy (+2.96%) and 72.64% AA F1 (+5.28%) over uniform noise. Masked paths allow `KnapsackDP` to evaluate remaining mass deficits against unassigned positions cleanly, whereas uniform categorical noise initializes with arbitrary intermediate masses that conflict with early reachability bounds.

Integration Steps and Scheduling: Table 3b examines reverse Euler steps $K \in \{5, 10, 20\}$. At $K = 5$, DFlowNovo achieves 58.40% accuracy at 410.2 spec/s, suitable for rapid screening. The default setting of $K = 20$ with cosine scheduling achieves 62.81% accuracy at 105.2 spec/s, offering a strong balance between accuracy and computational cost. Cosine scheduling slightly outperforms linear scheduling (+0.71% accuracy) by providing finer step resolution near $t = 1$ as residue assignments solidify.

4.5. Precision–Coverage Trade-off and Calibration

In proteomics pipelines, identifications are typically filtered by score thresholds to control false discovery rates. Figure 3 shows empirical precision–coverage curves on Nine-Species. DFlowNovo achieves high residue precision ($\text{AUC}_{\text{aa}} = 0.941$ vs. 0.905 for InstaNovo v1.2, 0.894 for InstaNovo+, and 0.817 for Casanovo), sustaining 98.54% precision at 50% coverage. In sequence accuracy, DFlowNovo achieves $\text{AUC}_{\text{pep}} = 0.859$ (vs. 0.845 for InstaNovo v1.2, 0.839 for InstaNovo+, and 0.746 for Casanovo), reaching 91.56% accuracy at 50% coverage. The smooth, monotonic profile indicates that ListNet scores are well calibrated for downstream FDR filtering.

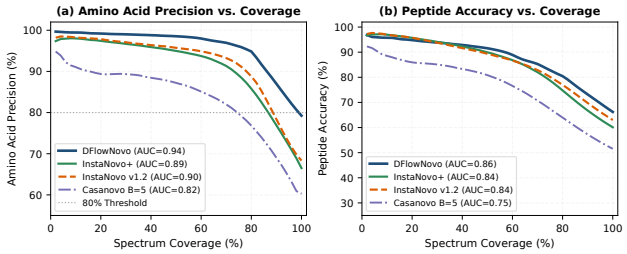


Figure 3. **Precision–Coverage Benchmark on Nine-Species.** Empirical curves on the full test split. (a) Residue-level precision ($\text{AUC}_{\text{aa}} = 0.941$ vs. 0.905 for InstaNovo v1.2, 0.894 for InstaNovo+, and 0.817 for Casanovo). (b) Peptide sequence accuracy ($\text{AUC}_{\text{pep}} = 0.859$ vs. 0.845 for InstaNovo v1.2, 0.839 for InstaNovo+, and 0.746 for Casanovo). DFlowNovo sustains $> 98\%$ residue precision across the top 50% coverage, confirming well-calibrated confidence scores for FDR control.

5. Conclusion and Discussion

We presented **DFlowNovo**, a non-autoregressive framework for *de novo* peptide sequencing based on Continuous-Time Markov Chain (CTMC) Discrete Flow Matching. By operating directly on discrete distributions, DFlowNovo avoids both sequential decoding bottlenecks and continuous-to-discrete projection mismatches. Through GPU-accelerated `KnapsackDP` reachability filtering, the model enforces precursor mass conservation in $\mathcal{O}(1)$ time per residue during generation. On the Nine-Species benchmark under MassIVE-KB pretraining, DFlowNovo achieves 62.81% Peptide Accuracy and 78.06% Amino Acid F1 (+9.62% over InstaNovo v1.2) at 105.2 spectra/s with 59.5M parameters.

Limitations and Future Work: Like most non-autoregressive sequence models, DFlowNovo relies on predicting sequence length prior to decoding, which we address via length hypothesis sampling. Extending dynamic programming reachability to combinatorial post-translational modifications with arbitrary variable mass shifts remains an open direction. More broadly, coupling discrete flow matching with dynamic programming reachability offers a practical way to generate discrete sequences subject to exact linear sum constraints.

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A. Implementation Details, Hyperparameters, and Hardware

A.1. Model Architecture Specifications

Table 4 summarizes architectural specifications for DFlowNovo. The entire model comprises 59,496,750 parameters (59.50M), partitioned between the Spectrum Transformer Encoder (16.29M, 27.4%), the Discrete Flow Decoder (42.85M, 72.0%), and the Precursor Length Predictor (0.36M, 0.6%).

Module	Hyperparameter	Value
Spectrum Encoder	Transformer Layers	6
	Hidden Dimension (d_{model})	512
	Attention Heads	8
	Feed-Forward Dimension (d_{ffn})	1536
	Attention Implementation	FlashAttention-2
	Max Input Centroided Peaks (P)	200
	Peak Embeddings	Dual Sinusoidal (Fragment + Complementary)
Discrete Flow Decoder	Transformer Blocks	6
	Hidden Dimension (d_{model})	512
	Attention Heads	8
	Feed-Forward Dimension (d_{ffn})	1536
	Feed-Forward Activation	SwiGLU
	Time & Condition Mechanism	AdaLN-Zero on $(t, M_{\text{prec}}, z, L)$
	Categorical Vocabulary Size (S)	32 (20 Canonical + 10 PTMs + [PAD] + [M]; 30 active classes)
Length Classifier	Architecture	2-Layer MLP on $h_{\text{[CLS]}}$
	Hidden Dimension	512
	Output Classes (L)	$1 \leq L \leq 50$ (50 classes)
Inference & Decoding	Reverse Euler Steps (K)	20 (Cosine Schedule)
	Detailed Balance Rate (η)	0.15 (Window $t \in [0.2, 0.8]$)
	Candidate Pool Size (C)	6 (Hybrid Beam)
	Greedy Decoding Temp (T)	0.0 (Candidate 1, MAP argmax)
	Stochastic Decoding Temp (T)	0.7 (Candidates 2–6, categorical sampling)
	Mass Grid Resolution (Δm)	0.02 Da (Tolerance $\tau = \pm 0.1$ Da, $W = 11$)

Table 4. **Hyperparameter specifications for DFlowNovo.**

A.2. Training Protocol and Optimization

DFlowNovo was trained on the MassIVE-KB repository ($N = 28{,}232{,}112$ human spectra) using PyTorch 2.5 and PyTorch Lightning. Optimization utilized AdamW with $\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-8}$, and weight decay of 10^{-4}. The learning rate followed a Warmup-Stable-Decay (WSD) schedule with warmup ratio 0.05, stable ratio 0.70, peak learning rate $\eta_{\max} = 2 \times 10^{-4}$, and minimum learning rate ratio 0.01. Training utilized mixed-precision (bfloat16) on NVIDIA H100 80GB GPUs with an effective batch size of 1,024 spectra.

A.3. Code and Data Availability

- **Source Code and Checkpoints:** The official implementation of DFlowNovo, training configurations, inference scripts, precomputed knapsack tables, and evaluation pipelines are openly available at https://github.com/joelinator/JoelResearch. Pretrained model weights are hosted on the Hugging Face Hub at https://huggingface.co/joelinator/dflow-novo-model.

- **MassIVE-KB Pretraining Corpus:** Tandem mass spectrometry spectra from the human MassIVE-KB dataset are publicly hosted by UCSD under MassIVE accession MSV000084075 (https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?accession=MSV000084075) and available as preprocessed Parquet shards on Hugging Face Datasets at https://huggingface.co/datasets/chrisagrams/massive_kb_v1_shuffled.

- **Nine-Species Benchmark Dataset:** The multi-species tandem mass spectrometry benchmark dataset ($N = 2{,}322{,}730$ spectra across nine organisms) is available via the PRIDE repository under accession

PXD004948 and hosted on Hugging Face Datasets at https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark.

B. CTMC Discrete Flow Matching Mathematical Foundations

In this section, we provide the concise mathematical foundations for Continuous-Time Markov Chain (CTMC) Discrete Flow Matching on probability simplices, formalizing the framework of Campbell et al. (Campbell et al., 2024).

B.1. Continuous-Time Master Equation and Rate Matrices

Let $\mathcal{X} = \{1, \ldots, S\}$ be a finite categorical state space of size $S = 21$ (20 amino acids + $[M]$). The time evolution of the marginal probability vector $\mathbf{p}_t \in \Delta^{S-1}$ under a continuous-time Markov chain is governed by:

$$\frac{d}{dt}\mathbf{p}_t = \mathbf{p}_t R_t \tag{17}$$

where $R_t \in \mathbb{R}^{S\times S}$ is the transition rate matrix satisfying $R_t(j \mid i) \geq 0$ for $i \neq j$, and diagonal elements $R_t(i \mid i) = -\sum_{j\neq i} R_t(j \mid i)$.

B.2. Conditional Jump-Minimal Dynamics

Given target token $y \in \mathcal{V}_{\text{AA}}$ at $t = 1$ and initial state $[M]$ at $t = 0$, the masked conditional probability path is:

$$p_{t|1}(x \mid y) = \big(1 - \kappa(t)\big)\delta_{[M]}(x) + \kappa(t)\delta_y(x) \tag{18}$$

where $\kappa(t)$ is a smooth schedule with $\kappa(0) = 0, \kappa(1) = 1$. Differentiating with respect to t:

$$\frac{d}{dt}p_{t|1}(x \mid y) = -\dot{\kappa}(t)\delta_{[M]}(x) + \dot{\kappa}(t)\delta_y(x) \tag{19}$$

Under jump-minimal transition dynamics, transitions occur directly from $[M]$ to y:

$$R_t(y \mid [M], y) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{20}$$

For any other pair (i, j) with $i \neq [M]$ or $j \neq y$, $R_t(j \mid i, y) = 0$.

B.3. Simulation-Free Marginal Vector Field

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the marginal transition rate from mask $[M]$ to residue $a \in \mathcal{V}_{\text{AA}}$ is:

$$R_t(a \mid [M], \mathcal{S}) = \sum_y R_t(a \mid [M], y)q(y \mid x = [M], \mathcal{S}) \tag{21}$$

$$= \frac{\dot{\kappa}(t)}{1 - \kappa(t)}\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S}) \tag{22}$$

where $\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S})$ is the neural posterior predicted by the discrete flow decoder. Training by minimizing cross-entropy over masked positions (Equation (10)) directly minimizes the KL divergence between the true velocity field and the parameterized model.

C. Combinatorial KnapsackDP Exactness and Complexity

C.1. Formulation and Exactness

Let $\mathcal{W} = \{\lfloor m(a)/\Delta m \rceil \mid a \in \mathcal{V}_{\text{AA}}\} \subset \mathbb{N}$ denote the set of integer-quantized amino acid masses on grid $\Delta m = 0.02\,\text{Da}$. Let $T[k, b] \in \{0, 1\}$ be the boolean dynamic programming table defined in Equation (11).

Proposition C.1 (Reachability Exactness)**.** *For any positive integer $k \leq K_{\max}$ and mass bin $b \leq B_{\max}$, $T[k, b] = 1$ if and only if there exists a multiset of amino acids $\{a_1, \ldots, a_k\} \subseteq \mathcal{V}_{AA}$ such that $\sum_{i=1}^{k} \lfloor m(a_i)/\Delta m \rceil = b$.*

Algorithm 2 Precomputed GPU `KnapsackDP` Table Construction

Input: Residue masses $\{m(a)\}_{a\in\mathcal{V}_{\text{AA}}}$, grid resolution $\Delta m = 0.02\,\text{Da}$, max mass $M_{\max} = 5000\,\text{Da}$, max length $K_{\max} = 50$, instrument tolerance $\tau = 0.1\,\text{Da}$.
Initialize bin count $B_{\max} \leftarrow \lfloor M_{\max}/\Delta m \rfloor = 250{,}000$.
Allocate boolean table $T \in \{0,1\}^{(K_{\max}+1)\times B_{\max}} \leftarrow 0$; set $T[0,0] \leftarrow 1$.
for $k = 1$ **to** $K_{\max}$ **do**
  for $a \in \mathcal{V}_{\text{AA}}$ **do**
    $w_a \leftarrow \lfloor m(a)/\Delta m \rceil$
    $T[k, w_a : B_{\max}] \leftarrow T[k, w_a : B_{\max}] \vee T[k-1, 0 : B_{\max} - w_a]$
  end for
end for
Window size $W \leftarrow 2\lfloor \tau/\Delta m \rfloor + 1 = 11$.
Apply GPU 1D max-pooling with kernel size W:
$T_{\text{pooled}}[k, b] \leftarrow \max_{|j-b|\le 5} T[k, j]$.
Output: Pooled reachability tensor T_{pooled} in GPU memory.

Proof. By mathematical induction on residue count k. For $k = 0$, $T[0,0] = 1$ and $T[0,b] = 0$ for all $b > 0$, which holds trivially. Assume the proposition holds for $k-1$. For step k:

$$T[k,b] = \bigvee_{a\in\mathcal{V}_{\text{AA}}} T[k-1, b-w_a] \tag{23}$$

$T[k,b] = 1 \iff \exists a \in \mathcal{V}_{\text{AA}}$ such that $T[k-1, b-w_a] = 1$. By the inductive hypothesis, $T[k-1,b-w_a] = 1 \iff \exists\{a_1,\ldots,a_{k-1}\}$ such that $\sum_{i=1}^{k-1} w_{a_i} = b - w_a$. Setting $a_k = a$ yields $\sum_{i=1}^{k} w_{a_i} = b$. Thus, $T[k,b] = 1$ if and only if the mass bin b is reachable by exactly k residues. □

C.2. Computational Complexity

- **Precomputation:** Computing table T requires $K_{\max} \times |\mathcal{V}_{\text{AA}}| \times B_{\max} = 50 \times 20 \times 250{,}000 = 2.5 \times 10^8$ boolean operations, executing in under 0.25 seconds on GPU once at initialization.
- **Inference Lookup:** At each flow step k, checking chemical admissibility (Equation (13)) consists of a single memory indexing operation $T_{\text{pooled}}[u-1, b]$ per residue candidate. This requires $\mathcal{O}(1)$ time per residue, introducing negligible overhead (< 1.8 ms per batch of 512 spectra).

D. Exhaustive Multi-Species Cross-Kingdom Generalization

To evaluate whether the learned continuous-time vector field generalizes across phylogenetic diversity without species-level overfitting, Table 5 reports the complete evaluation of DFlowNovo across all 2,322,730 spectra of the Nine-Species dataset. Pretrained exclusively on human MassIVE-KB spectra, non-human species represent zero-shot cross-organism evaluation.

Species Display Name	Taxonomic Group	Spectrum Count	Peptide Acc. (%)	Mass Match (%)	AA F1 (%)	AA Precision (%)	AA Recall (%)	Length Acc. (%)
Mus musculus (Mouse)	Mammalia	25,963	48.06	51.94	75.02	75.35	74.69	76.86
Homo sapiens (Human)	Mammalia	43,416	65.14	67.86	78.60	78.76	78.44	84.98
Candidatus endoloripes	Bacteria	79,144	48.98	52.57	71.18	71.46	70.91	75.09
Vigna mungo (Black gram)	Plantae	107,955	69.15	73.47	86.96	87.06	86.86	88.08
Saccharomyces cerevisiae (Yeast)	Fungi	104,163	66.44	68.24	81.16	81.28	81.04	84.77
Solanum lycopersicum (Tomato)	Plantae	177,721	69.13	71.94	81.93	82.04	81.82	86.81
Apis mellifera (Honeybee)	Insecta	192,397	55.01	58.75	72.12	72.31	71.93	78.69
Methanosarcina mazei	Archaea	262,739	62.59	66.57	78.40	78.47	78.33	83.04
Bacillus subtilis	Bacteria	1,329,232	65.83	68.78	77.72	77.89	77.56	84.29
Macro-Average / Total	**6 Biological Kingdoms**	**2,322,730**	**61.15**	**64.46**	**78.12**	**78.29**	**77.95**	**82.25**

Table 5. **Per-species performance breakdown across all nine species (N = 2,322,730 spectra total).** Across all nine organisms, DFlowNovo recovers 224,249 unique peptides with a macro-average Amino Acid F1 of 78.12% and Peptide Accuracy of 61.15%.